



**Canopus**  
The second-brightest star in the entire sky, Canopus is a white supergiant, 310 light-years away. At magnitude  $-0.62$ , it is easily visible to the naked eye. It is a truly brilliant star—placed next to the Sun, it would outshine it 14,000 times.

## SOUTHERN LATITUDES

Looking north, the spring constellations Aquarius, Pisces, and Pegasus are moving toward the western horizon as the summer stars rise in the east. The upside-down figure of Orion, the hunter, announces that summer is almost here. It is flanked by Taurus to its lower left and Canis Major to the right. Below is the constellation Gemini, the twins. The two bright stars that mark their heads, Castor and Pollux, are near the northeastern horizon. Their legs point upward to Orion.

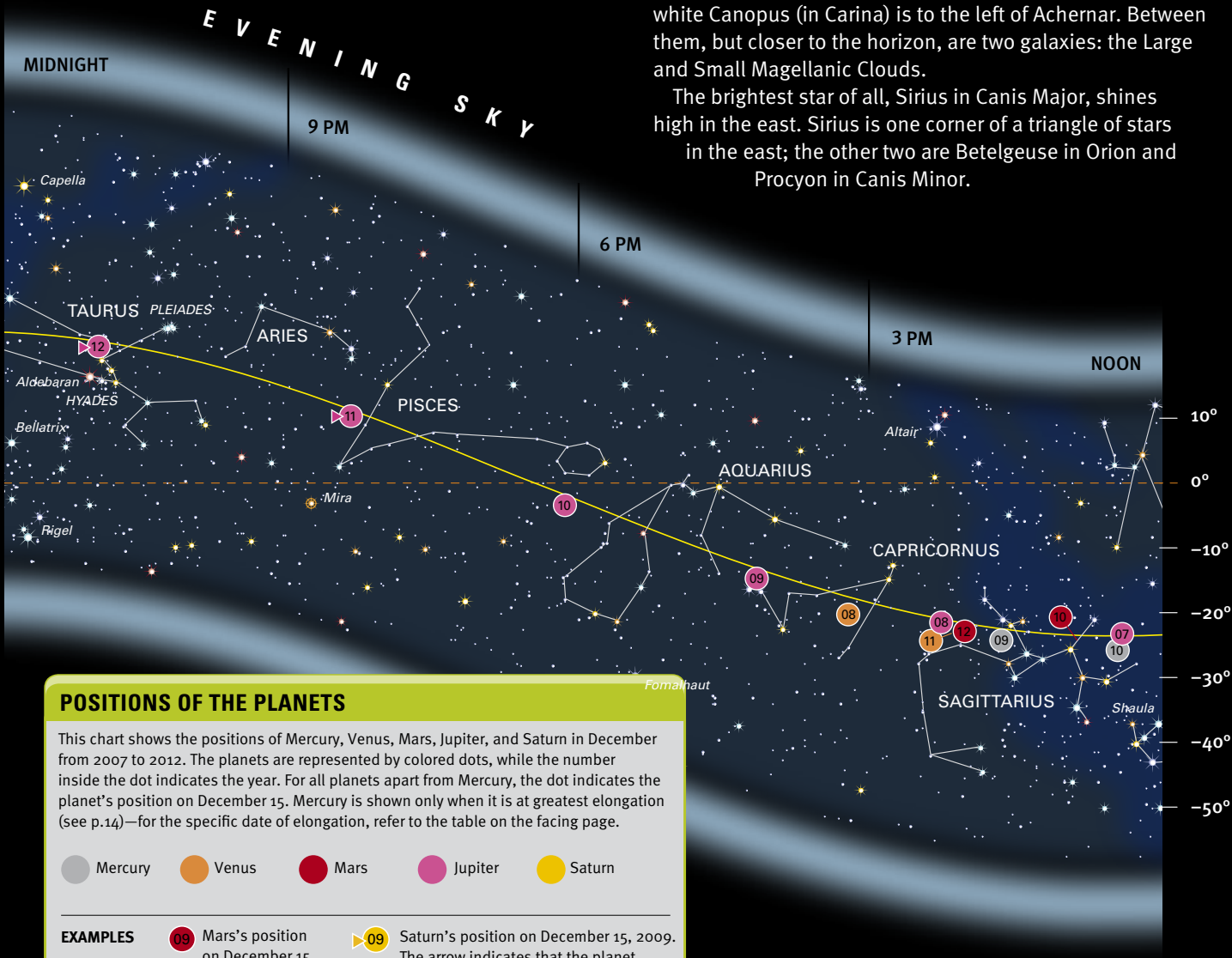
Taurus is the front portion of a bull, which, from southern latitudes, is seen facing east, the direction it has come from. Its shoulders and head are close to the horizon, and its forelegs point up to the sky overhead. The Pleiades (M45), one of the finest open clusters in the sky, marks the bull's shoulder. Six of its stars are discernible with the naked eye; the sharp-sighted may see seven, which gives rise to the cluster's alternative name, the Seven Sisters. Many more are visible with optical aid. The bull's face is formed of a second, V-shaped cluster, the Hyades. More than a dozen of its stars are visible to the naked eye. The unrelated red star Aldebaran represents one eye.

Eridanus, the river, is overhead. Its meandering path is traced from Orion's bright star, Rigel, a blue supergiant of magnitude 0.18. It flows overhead, then toward the southern horizon, stopping at Achernar at right of due south. Brilliant-white Canopus (in Carina) is to the left of Achernar. Between them, but closer to the horizon, are two galaxies: the Large and Small Magellanic Clouds.

The brightest star of all, Sirius in Canis Major, shines high in the east. Sirius is one corner of a triangle of stars in the east; the other two are Betelgeuse in Orion and Procyon in Canis Minor.



**Hyades and Pleiades**  
The V-shaped group of stars below and left of center is the Hyades star cluster representing the face of Taurus. The brightest star in the face is Aldebaran, which is not part of the cluster. The knot of stars at center right is the Pleiades cluster.



### POSITIONS OF THE PLANETS

This chart shows the positions of Mercury, Venus, Mars, Jupiter, and Saturn in December from 2007 to 2012. The planets are represented by colored dots, while the number inside the dot indicates the year. For all planets apart from Mercury, the dot indicates the planet's position on December 15. Mercury is shown only when it is at greatest elongation (see p.14)—for the specific date of elongation, refer to the table on the facing page.

Mercury Venus Mars Jupiter Saturn

**EXAMPLES** 09 Mars's position on December 15, 2009. 09 Saturn's position on December 15, 2009. The arrow indicates that the planet is in retrograde motion (see p.14)